The flow value of being alive אis matched to a value of a statistical life (VSL) of 80 times average annual income (Viscusi and Aldy, 2003).12 The parameter b, which sets the marginal utility of the first unit of drugs, is chosen to match a drug participation rate of 8.90 percent, the share of respondents aged 26 and over in the 2024 NSDUH with a drug use disorder of any severity. The weight on drugs θ is set to match the aggregate drug spending over the aggregate income of users, 16.0 percent, from retail spending in Midgette et al. (2019) and the incomes of users in the 2024 NSDUH. The level of food assistance ¯c is set so that total food assistance is 0.42 percent of total labour income, the ratio of SNAP outlays to gross domestic product (GDP).

The mortality exponent κϕd is set to match the average income of users, 0.74 of the population average in the 2024 NSDUH, because the death risk is what decides how far down the income distribution use sits, as Section 3.3 shows.

3.3 Who uses drugs

Drugs in the model are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. If a SUD had no consequences, that would be the only force operating, and drug use would concentrate at the top: with b refitted so that participation stays at 8.90 percent, the average user would then earn 3.51 times the population average. What follows shows that it is the uninsurable risk of dying that reverses this gradient.13

Figure 1 scales each of the two channels through which a SUD reaches old age, mortality and unem- ployment, from nothing to 200 percent of its calibrated value, with the other switched off. It scales κϕd -1

11The annual discount factor here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon.

12No household has a negative value of life in period 1, so death is never an optimal choice. 13A household can buy life insurance, but it protects its descendants, not the household itself or the value of its own life.

Table 1: Externally set parameters

Parameter Description Source or rationale Value

β Discount factor over a thirty-year period 0.995 annually, compounded 0.86 σ Risk aversion Standard; Attanasio and Weber (2010) 2 ¯ϕ Survival over a period, no SUD US life table 2010, Arias (2014), ages 26–55 0.93 πd Chance that use starts a SUD NSDUH 2024: SUD rate over participation 0.41 σy Dispersion of log productivity SCF 2022: bottom half holds 17.2% of earnings 0.95 u0 Unemployment rate in period 2, no SUD BLS CPS, average 2014–2024 0.049 u1 Unemployment rate in period 2 with a SUD NSDUH 2024, ages 26 and over 0.16 ϱ Food assistance taper SNAP benefit reduction rate 0.30 pv Price of

a unit of drugs Normalisation 1.00

Table 2: Internally calibrated parameters

Parameter Target moment Data Model Value

א VSL over income 80 80.0 8.64 b Drug participation, NSDUH 2024 8.9% 8.9% 0.032 θ Drug spending over users' income 16.0% 16.0% 0.22 κϕd Users' income over the average, NSDUH 2024 0.74 0.74 3.56 ¯c Food assistance over labour income 0.42% 0.42% 0.10

for the mortality channel and u1 − u0 for the unemployment channel. At every point b is refitted so that participation stays at its calibrated value: the share of households that use is the same everywhere on the figure, and only who uses changes.

The left panel shows the model, which allows households to save. The mortality channel alone takes the average user from 3.51 times the population average down to 0.64 at its calibrated strength, crossing the average at 73 percent of that strength, and to 0.27 at 200 percent.14 In contrast, the unemployment channel alone leaves the average user at 3.51 times the population average even at 200 percent: on its own it does not change who uses at all.

To understand why the risk of dying reverses the gradient and the risk of losing one's job does not, I turn to the right panel, which switches saving off and leaves every parameter at its calibrated value, so households cannot hold precautionary savings against job loss. Without savings, the unemployment channel also reverses the gradient: on its own it takes the average user to 0.75 times the population average at its calibrated strength, and to 0.27 at 200 percent.

Who uses is decided by a race between two things that change with income: the cost of acquiring drugs and the penalty they inflict. The utility cost of the drug declines with income because a household with a higher income gives up less marginal utility for each dollar it spends on drugs, so buying v units costs a rich household little in terms of utility relative to a poor household.

The penalty inflicted by drugs does not rise with income when it can be partially insured against, but does rise when it cannot be. Unemployment can be partly insured against through saving: a richer household carries more savings into a spell out of work, so the loss is proportional to what it has, and thus no greater in utility terms for a rich household than for a poor one. Death, on the other hand, cannot be insured against. It destroys the whole continuation value: the consumption of a life still to be lived, and א, the value of being alive at all. The loss therefore grows with income, deterring high-income households and not low-income ones. Without the ability to save, losing a job takes a household down to ¯c whatever it earned, so that loss too grows with income, which is why in the right panel the unemployment channel reverses the gradient as well.

14With both channels at their calibrated strength, the average user earns 0.74 times the population average, as in the data.

0 50 100 150 200 Strength of the channel, percent of calibrated

0.0

0.5

1.0

1.5

2.0

2.5

3.0

3.5

Mean income of users / population mean

calibrated

users poorer than average

0.27

3.51 3.51

With saving (the model)

Mortality risk of a SUD Unemployment risk of a SUD

0 50 100 150 200 Strength of the channel, percent of calibrated

calibrated

users poorer than average

0.27 0.27

3.51

Saving switched off

Figure 1: What it takes to reverse the income gradient in drug use

Notes: each curve scales one channel with the other switched off. Left: the model. Right: the same parameters with saving switched off. The vertical axis is average income of users over the population average, and b is recalibrated at every point so that participation is the calibrated 8.90 percent. The vertical dotted line marks the calibrated strength.

4 The quantitative model

This section describes the model the paper's numbers come from. It keeps the preferences of the stylized model of Section 3 and the two risks a SUD brings there, and adds what that

model leaves out: hous- ing, homelessness and vouchers; annual periods; unemployment insurance; income risk during working age; retirement; progressive income taxes; interest on savings; a drug-use decision in every year; recovery and treatment; SMI; a higher risk of dying, worse employment prospects and a lower chance of recovering while homeless; children who live with their parents; a drug test that misses some users; a government budget held at its benchmark net revenue; and an endogenous drug price. Notation follows Section 3 wherever the two models share an object: θ is the drug weight, σ the risk aversion, b the Stone-Geary shifter, אthe flow value of being alive, πd the chance that use starts a SUD, κϕd the mortality multiplier for the disorder, pv the drug price and β the discount factor. The values are in Section 5.

4.1 Demographics and the state and choice sets

At model age $j = 1$, a child becomes an adult, leaves its parent's home, joins the workforce and forms its own household, with a child of its own. The household lives at most J annual periods, works while $j < JR$ and is retired from JR. Its child lives with it for the first JK years and then leaves to form its own household in turn.

Each period a cohort of unit mass enters at $j = 1$, made up of the children who leave home that period, at their parent's death if that comes before JK and at JK otherwise (equation (29)). It replaces the mass of the households who died, so the mass of households is constant in a stationary equilibrium. The individual state is age j together with

$$s = (a, y, e, d, m, \eta) \in R+ \times Y(j) \times \{1, 2, 3\} \times \{0, 1, 2\} \times \{0, 1\} \times \{0, 1, 2\}, \quad (9)$$

where a denotes assets, y productivity, e employment status, d the SUD, m SMI and η the household's standing with the voucher programme. The SUD takes three values because the parent's valuation of its child's prospects depends on the history of the disorder and not only on its present (Section 4.10): $d = 0$ denotes a household that has never had the disorder, $d = 1$ refers to one that had it and has recovered, and $d = 2$ stands for one that has it now.15 The voucher state is $\eta = 0$ for a household without a voucher,

15Section 3 wrote $d \in \{0, 1\}$ for the disorder absent and present; here $d = 1$ is a household that has recovered, since the quantitative model keeps the history.

$\eta = 1$ for one that holds one, and $\eta = 2$ for one excluded from the programme after failing Treatment First's condition (Section 4.9).

Each period the household chooses the bundle

$$\text{h, v, } \tau, \text{ a}'$$

$$\in \{h1, h2, h3, h4\} \times R+ \times \{0, 1\} \times R+, \quad (10)$$

where h is the dwelling, of which h1 is a shelter, v is drug consumption, τ is whether or not the household is in treatment for the SUD, available only to a household that has the disorder, and a′ is its next period's assets. Labour is inelastically supplied when employed.

4.2 Preferences

Flow utility is the stylized model's, with housing in the bundle and the child present for

the first JK years. Consumption c and housing h are shared through the equivalence scale ξ(j), which counts two members while the child is at home and one after. The drug enters the parent's utility alone, so the parent and the child hold the bundles

XP =

��c ξ(j)

�1 − γ�h ξ(j)

�γ�1 − θ (b + v)θ, Xc =

��c ξ(j)

�1 − γ�h ξ(j)

�γ�1 − θ bθ, (11)

and period utility is

u

� c, h, v, τ

� = U(XP) + ωc (ξ(j) − 1) U(Xc) + א− τ ¯τ u, U(X) = X1 − σ

1 − σ . (12)

γ is housing's share of the non-drug bundle and θ the drug's share of the whole, σ is the coefficient of relative risk aversion, אis a flow value of being alive, counted once per household, and ¯τ u is the utility cost of a year in treatment, which is described in Section 4.6. The child's term is its own flow utility, counted as ξ(j) − 1 of the household and weighted by the parent at ωc. The child gets none of the drug, so a parent's drug spending reaches it through a reduction in the consumption and housing they share.

4.3 Income, taxes and transfers